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Best practice recommendations for bioinformatics approaches applied to high-throughput sequencing for rare disease and cancer diagnosis within the UK National Health Service

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ABSTRACT

Background Establishing best practice recommendations helps to increase consistency, equity and innovation in clinical genomics services. Bioinformatics approaches are a core component of clinical genomics services that use high-throughput genomic sequencing applied in the diagnosis of rare disorders and cancer. While a broad range of international recommendations exist for genomic diagnostic testing and genetic variant classification, the current UK-specific best practice recommendations for bioinformatics approaches applied in this context are outdated.

Methods We assembled a team of bioinformaticians and scientists with diverse expertise in rare disease and cancer genomics applied in clinical diagnostics within the UK National Health Service. Through structured discussion, polls and surveys, we developed an updated set of best practice recommendations for bioinformatics approaches applied to high-throughput genomic sequencing in clinical genomic testing.

Results We provide best practice recommendations across the spectrum of activities within a clinical genomics bioinformatics pipeline, including quality control, primary, secondary and tertiary analysis approaches and shared knowledge bases. We also comment on issues related to software development and maintenance. The recommendations can be applied to multiple sequencing technologies and encompass both targeted and whole genome sequencing approaches applied to germline and tumour DNA samples.

Conclusion The best practice recommendations outlined in this study provide a national framework for adoption and innovation of bioinformatics approaches across diverse clinical genomic testing strategies in the UK National Health Service.

INTRODUCTION

The application of genomic sequencing approaches to assist in the diagnosis and management of rare disorders and cancer has proven clinical utility across diverse areas.^{1–5} Examples of clinical utility include informed clinical diagnosis, wider family testing and support with family planning, informed prognosis and treatment decisions, clinical trial eligibility and confirmation of histological findings.

WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ Best practices for bioinformatics approaches applied to clinical genomics in the UK were previously published in 2016.
- ⇒ Given the evolving nature of diagnostic genomic strategies, bioinformatics and high-throughput sequencing approaches, an update to best practice recommendations adopted across bioinformatics service providers in the UK is required to ensure consistency and quality.

WHAT THIS STUDY ADDS

- ⇒ We source diverse expert viewpoints and provide an updated consensus opinion on best practice bioinformatics approaches applied in clinical genomics, including quality control, primary, secondary and tertiary bioinformatics analysis, shared knowledge bases and software development.
- ⇒ The best practice recommendations proposed here are applicable across sequencing technologies and apply to diverse clinical scenarios, including rare disease and cancer diagnostic strategies.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ We anticipate that this guidance will be considered and implemented across broad areas of clinical genomics applied within the UK National Health Service, and provide a transparent framework for consideration in the development and innovation of clinically accredited bioinformatics services.
- ⇒ We expect this guidance to increase reproducibility, equity and innovation within national rare disease and cancer clinical genomics strategies, and while developed for the UK context, to serve as a model for other countries seeking to update their clinical bioinformatics standards and accreditation frameworks.

The UK and the National Health Service (NHS) have been at the forefront of the development of national genomic research initiatives integrated with

diagnostic laboratories,^{3 5–9} and the innovation and adoption of genomic technologies in routine diagnostic pathways.^{10 11} There remain opportunities to evolve and improve diagnostic genomic sequencing, reinforced by recent and continual discovery of new genomic disorders,^{12–16} new types of genomic variants contributing to genomic disorders^{17–21} and increasing access to precision cancer therapies.^{22 23}

A number of approved guidelines and recommendations for aspects of the diagnostic genomic sequencing pathway exist.^{24–37} Best practices have been reinforced and repurposed for use in UK healthcare services and are reviewed regularly.³⁸ To date, however, recommendations for best practices of bioinformatics approaches for clinical genomic high-throughput sequencing adopted in the UK remain largely outdated,³⁹ although recent guidance from other international communities continues to emerge.^{33 36 37} Updating guidance for adoption of bioinformatics in clinical genomics is especially timely with the rapidly evolving nature of genomic testing strategies, sequencing methodologies and bioinformatics software.

In this study, we develop recommended best practices for bioinformatics approaches applied to clinical genomic high-throughput sequencing within the UK NHS, where medical device regulations are distinct from those applied in the European Union and there is a requirement within the national genomic services for the devolved nations (England, Wales, Scotland and Northern Ireland) for equitable access to clinical genomic testing. An additional challenge is consistent quality standards where there are geographically distributed genomic diagnostic laboratories within nations, for example, the regional NHS England Genomic Laboratory Hubs. Best practice recommendations form a critical resource for formal clinical accreditation for genomic diagnostic services and for training; however, the key reference for bioinformatics practice remains The Association for Clinical Genomic Science (ACGS) recommendations published in 2016. Here, through an ACGS working group, assembled to represent national genomic services in England, Wales and Scotland, we update the 2016 recommendations and conduct a comprehensive review of existing international publications. The major aim of this work is to expand and reinforce best practice recommendations and provide an end-to-end innovation framework for bioinformatics approaches applied to clinical genomics within the UK NHS. We address issues relating to the validation of approaches adopted for primary (library preparation and sequence generation), secondary (sequencing read alignment and variant detection) and tertiary (variant annotation, prioritisation and classification) stages of bioinformatics pipelines (see online supplemental table 1 for further definition), along with comments on best practices for shared knowledge bases, software development and maintenance in the context of clinical genomic sequencing. It is important to note that these recommendations do not represent all aspects of clinical bioinformatics and as such should be taken as an addition to the literature. Our findings provide a nationally coordinated framework for bioinformatics applied to clinical genomics within the UK and provide a model for adoption in other countries and healthcare systems.

METHODS

Assembly of representative expert team to create best practice recommendations

We assembled a team of bioinformaticians and scientists with diverse expertise in rare disease and cancer genomics, including representatives from five of seven NHS England Genomic

Laboratory Hubs, the All Wales Medical Genomics Service and Genomics England. Individuals who formed this working group were nominated through an open call to national leaders in bioinformatics applied to clinical genomics in the UK NHS (online supplemental figure 1). The experience of the working group, listed as authorship to this manuscript, represents most clinical services for rare disease and cancer that use high-throughput sequencing within the UK NHS.

PROCESS FOR DRAFTING AND REFINING BEST PRACTICE RECOMMENDATIONS

A summary workflow for drafting and refining the best practice recommendations is provided in online supplemental figure 1. Information and opinions were collected and documented through structured group discussions, surveys and polls. Examples of survey questions and discussion points are included in online supplemental table 2. A writing group, consisting of four scientists and bioinformaticians (JE, rare disease genomics; AP, translational bioinformatics; EW, cancer bioinformatics and JWA, lead bioinformatician) drafted the consensus recommendations after discussions and written feedback from the working group. Recommendations and the draft manuscript were circulated to all working group members prior to additional discussions and refinement of the guidance. Feedback from these wider reviews was incorporated into the tables of recommendations and the manuscript. Separate discussion sessions were conducted for distinct sections of the guidance, as partitioned in the results section, with final review of recommendations and decisions led by the writing group. All working group members received written communication of the best practice recommendations and gave their written approval. The recommendations were further reviewed by lead bioinformaticians across laboratories providing NHS genomic medicine services and were open for review and consultation by ACGS stakeholders and ACGS members (online supplemental figure 1). Any written feedback received was considered and integrated into the tables of recommendations and the manuscript, representing a nationally coordinated effort to create and refine best practice recommendations.

Comparisons of best practice recommendations developed by this team with other literature

After development and finalisation of the best practice recommendations included in the tables of recommendation described in this manuscript (tables 1–6), we compared our findings with those reported in other literature.^{24 25 33 36 37 39} We assigned one of four possible categories to each recommendation: ‘aligns’, guidance is identical/similar to that published previously and noted areas where the principle of the recommendation aligns but there are small differences in the exact recommendation; ‘expands’, guidance is similar but there is extended scope or detail provided in this manuscript; ‘discrepancy’, guidance is different to that provided elsewhere and ‘novel’, guidance is newly provided in this manuscript compared with that provided in the comparison literature.

Definition of terminology used throughout best practice recommendations

Broadly, we define best practice recommendations into two groups: *shall*, best practices that are expected, as a minimum, to be in place and *should*, best practices that are beneficial and opportunities to enforce them are expected to be sought, where possible. The development of a bioinformatics pipeline requires coordination and agreement with multidisciplinary teams, and

Table 1 Summary of best practice recommendations associated with primary analysis and QC that *shall* be adhered to

Category	Rationale	Best practice recommendation
Sequencing metrics	Ensuring that sequencing data are of sufficient quality, yield and evenness is critical for establishing that samples can be surveyed appropriately.	▶ Minimum requirements <i>shall</i> be in place for sequencing metrics at a sample level, including sequencing yield and coverage homogeneity/horizontal coverage. ▶ The precise QC approaches that <i>shall</i> be applied are dependent on the exact testing approach, and for tumour datasets <i>shall</i> include assessment of tumour purity and DNA degradation. Testing strategies that focus on specific hotspots and/or specific variants are expected to have QC requirements that are appropriate for the scope of the test. ▶ Monitoring of QC trends <i>should</i> be performed at the sample, sequencing run and sequencing lane levels, including, but not limited to, sequencing yield, the percentage of high-quality sequencing bases and average insert sizes.
Cross-contamination	Samples can become contaminated during sample collection and/or preparation for sequencing. Cross-contamination can lead to errors in variant detection and issues with other QC checks.	<i>Germline</i> ▶ In cases where a significant level of cross-contamination is identified, the laboratory <i>should</i> repeat the generation of sequencing data. In exceptional scenarios, it may be appropriate to use contaminated germline samples in the analysis (eg, contaminated at source and not possible to obtain another sample); in this scenario, the results <i>should</i> be considered out of scope of accreditation. ▶ Appropriate thresholds and toolkits to detect contamination <i>should</i> be benchmarked using known examples and in silico methods, where required, to understand impact on variant detection accuracy. ▶ Approaches <i>should</i> consider the impact of cross-contamination on de novo variant detection and analysis, and the performance of approaches across diverse genetic ancestries and in the presence of contamination from related individuals. <i>Tumour</i> ▶ Every effort <i>should</i> be made to benchmark appropriate approaches and thresholds for QC, where possible. ▶ Approaches to detect contamination of tumour samples in germline samples from the same individual <i>shall</i> be benchmarked and applied where both sequencing datasets are available.
Generated from expected laboratory sample	The accuracy and utility of applied testing approaches can be impacted by incorrect utilisation of sex karyotypes, and incorrect assignment of de novo variants and disease segregation patterns.	<i>Sex karyotype checks</i> ▶ Where sequencing reads from the testing strategy align to sex chromosomes, the QC approaches performed <i>should</i> assess concordance between reported sex and genetically inferred sex karyotypes. ▶ Strategies for mitigation <i>shall</i> be in place for mismatches identified between reported sex and genetically inferred sex karyotypes. <i>Familial relationships</i> ▶ Where sequencing data are available for multiple family members, and these data will be used to assist in variant detection and/or prioritisation strategies, bioinformatics approaches <i>shall</i> ensure that genetically inferred relationships between family members are consistent with reported relationships. ▶ At a minimum, identity-by-descent checks <i>shall</i> be used for available samples, to assess concordance with all reported relationships within the pedigree. Genetic concordance with reported relationships <i>should</i> be evaluated using benchmarked thresholds for familial relationships, as elaborated in online supplemental table 4. <i>Cohort-level identity checks</i> ▶ The capability to perform sequence identity checks across cohorts, including library preparation batch, sequencing flow cell and other informative groups <i>shall</i> be considered for incorporation. This approach can identify identical genetic samples as a result of sample mix-ups that are not captured by sex karyotype or familial relationship checks. <i>Paired tumour-normal samples</i> ▶ Where sequencing data are available for tumour and germline samples, the QC approaches <i>shall</i> ensure they are from the same individual, with some exceptions, for example, individuals who have undergone bone marrow transplantation.
Variant type-specific QC	The quality and consistency of the genomic sequencing data generated may differentially impact the detection of different types of genomic variants.	▶ The testing approach <i>should</i> benchmark the expected range of the number of variants in different classes (eg, single nucleotide variants, CNV) and provide appropriate warnings when samples fall outside of expected ranges. Warning thresholds <i>shall</i> be cognisant of scenarios such as hypermutated tumours, which invalidate this type of benchmark. ▶ Strategies <i>should</i> be in place to highlight samples that are more likely to have poor quality variant detection for certain classes of variants and <i>should</i> be appropriate for individuals across diverse genetic ancestries. ▶ Approaches that are undertaken to QC individual genomic variants <i>should</i> be benchmarked and justified.

Continued

Table 1 Continued

Category	Rationale	Best practice recommendation
Clinically relevant and/or incidental findings identified through QC process	QC processes may identify findings that are appropriate for clinical reporting and/or require clinical input to decide best course of action. QC approaches may also lead to incidental findings, including unexpected familial relationships and uniparental disomies (further elaborated in table 3).	► The QC processes put in place <i>shall</i> focus on the primary purpose of ensuring appropriateness of sequencing and/or metadata to limit inaccuracies in the applied genomic sequencing testing approach. ► Strategies <i>shall</i> be in place that enable appropriate evaluation and handling of cases that are detected from QC evaluations.
Of note, several QC metrics presented here require secondary analysis approaches to generate the data required for their assessment. Further definition of some metrics and details relating to their calculation are provided in table 2 . QC, quality control.		

throughout these best practice recommendations, we refer to *appropriate stakeholders*. For clarity, while *appropriate stakeholders* may be specific to testing strategies, examples include Health and Care Professions Council (HCPC) registered clinical genomic scientists, technical staff involved in development and delivery of services, genomic counsellors and clinicians that refer patients for diagnostic testing.

RESULTS

General considerations

The purpose of these best practice recommendations is to provide a framework for the innovation, upgrade and validation of bioinformatics software and/or approaches that are adopted in clinical genomics. The guidance is intended for UK-based diagnostic genomic sequencing services, although it may be appropriate for consideration in other countries and healthcare systems. Apart from exceptional circumstances, the general philosophy that should be taken is that the software being validated ‘should not perform worse than previous standard-of-care, and in most cases should provide advantages beyond previous testing strategies’. It is expected that a holistic view is taken when considering new developments and validations, and reduced performance in one area should be balanced against increased performance in others.

Examples of exceptional circumstances may include, but are not limited to, updates that are required due to security risks and/or updates that provide exceptional advantages in one area that outweigh loss in performance in other areas. In such scenarios, consideration *should* be given to the breadth and utility of other methodologies available in the NHS test directory to mitigate any alterations to current scope and utility. In addition, consideration *should* be given to opportunities enabled by software upgrades to expand beyond current approaches and utilities.

PRIMARY ANALYSIS AND QUALITY CONTROL

Primary analysis broadly refers to the generation of appropriate sequencing datasets for clinical high-throughput sequencing to be used in downstream secondary and tertiary processes (online supplemental table 1).²⁴ Quality control (QC) and assurance are key aspects of bioinformatics pipelines applied in clinical genomics. Appropriate QC *shall* be performed for all stages of the bioinformatics pipeline, including checks which can be performed during primary analysis (eg, assessment of some sequencing metrics), and after secondary analysis but relate to the generation of appropriate datasets (eg, sequencing read alignment, data consistency with key reported information and variant detection). Checkpoints for QC that *should* be applied will ensure that the sequencing data generated is appropriate for clinical analysis, is generated from the expected laboratory samples and is consistent with available information for the individual/family.

[Table 1](#) summarises agreed rationale and guidance for best practice recommendations that *shall* be adhered to for primary analysis and QC, although it should be noted that several of the QC metrics included require the completion of secondary analysis approaches prior to their calculation and assessments. These best practice recommendations do not seek to give prescriptive guidance on the exact thresholds used for QC, but aim to provide a general overview of the minimum requirements that *shall* be in place to ensure appropriateness of the sequencing datasets for clinical analysis. An example of appropriate thresholds for whole genome germline and tumour DNA sequencing approaches is presented in [table 2](#).

It is possible that clinically reportable events may be detectable through metrics collected for QC and not routinely identified through other secondary analysis approaches, for example, uniparental disomy (UPD).⁴⁰ Opportunities to include surveillance and reporting of these events *should* be considered where appropriate, with selected examples provided in [table 3](#).

Other aspects related to suitable file formats and the retention periods for specific files created during bioinformatics approaches applied in clinical genomic testing are discussed elsewhere and these standards/processes *shall* be adhered to.^{26 41} The British Society for Genomic Medicine recommendations for file retention⁴¹ provide detailed and nuanced recommendations and, in summary, state that aligned read file formats (eg, binary alignment map and/or compressed reference-orientated alignment map) are to be retained for 5 years as ‘working documents’, and that genetic variants and any associated genetic variant classifications should be retained long term to facilitate allele frequency calculations and inform further variant interpretation.

Secondary analysis approaches

Secondary analysis approaches broadly include sequencing read alignment, variant detection and related QC approaches applied in the pipeline (online supplemental table 1).²⁴ There are a number of considerations that impact the exact strategy adopted for secondary analysis, including: type of sequencing approach, for example, genome or targeted; sequencing technology; reference genome used; nature of test, for example, germline or somatic; types of genomic variants considered and other factors that are also within the scope of the testing strategy, for example, detection of epigenomic modifications.

When alterations are made to the approaches used for secondary analysis, the general considerations that *shall* be included are: (1) the impact on performance to detect genomic variants; (2) the impact on breadth/suitability of the testing strategy, for example, sequencing coverage in key areas for surveillance; (3) the impact on precision of downstream processes, for example, variant filtering and interpretation (further elaborated in section ‘Tertiary analysis approaches’) and (4) the impact on associated

Table 2 An example of minimum requirements* for quality control and quality assurance of genomic sequencing approaches for clinical diagnosis using short-read WGS technologies

Metric	Description	Suggested thresholds
<i>Sequencing data from germline samples</i>		
Sequencing yield	The total number of high-quality bases generated for each sample, as measured using a Phred-scaled quality score threshold of Q30, or equivalent. <i>Measurements should remove bases in duplicate reads and should not consider bases in clipped regions of reads or in overlapping regions between read pairs.</i>	Whole genome germline short-read DNA sequencing datasets with <85 Gb of Q30 bases <i>should be rejected</i> .
Coverage homogeneity/horizontal coverage	The percentage of nucleotides within the target region of the test and/or data generation region that obtain a predefined level of sequencing depth. <i>Measurements should not include duplicate reads, should not consider bases in clipped regions of reads or in overlapping regions between read pairs and should be limited to reads with high mapping quality (eg, Mapping Quality Scores (MAPQ) >10).</i>	Whole genome germline short-read DNA sequencing datasets with <95% of target positions covered at 15× or greater <i>should be rejected</i> .
Mean coverage	The level of sequencing coverage achieved (in line with requirements specified for <i>coverage homogeneity</i>), on average, for nucleotides included in the target region of the test and/or data generation region.	Whole genome germline short-read DNA sequencing datasets with <30× mean sequencing coverage, per base within target region, <i>should be rejected</i> .
Cross-contamination between samples	The relative proportion of cross-contamination estimated for the sample.	Whole genome germline short-read DNA sequencing datasets with >3% cross-contamination <i>should be rejected</i> .
Sample identity check	Where available, methods to genetically infer sex karyotypes <i>should be applied</i> , for example, relative coverage of sex chromosomes and/or coverage of the <i>SRY</i> gene <i>should be checked</i> against reported sex for the referred individual. Where familial methods for variant analysis are applied, genetic relatedness of referred family members <i>should be checked</i> for consistency with reported relationships.	Sample identity queries <i>should be escalated</i> to identify an appropriate outcome.
<i>Sequencing data from tumour samples</i>		
Sequencing yield	The total number of high-quality bases generated for each sample, as measured using a Phred-scaled quality score threshold of Q30, or equivalent. <i>Measurements should remove bases in duplicate reads and should not consider bases in clipped regions of reads or in overlapping regions between read pairs.</i> <i>Any warnings created by the bioinformatics pipeline may also consider reporting calculated level of tumour purity.</i>	Whole genome tumour short-read DNA sequencing datasets with <210 Gb of Q30 bases <i>should be rejected</i> .
Cross-patient contamination in tumour samples and sample swaps	The proportion of cross-patient contamination for tumour samples, through assessment of tumour and germline samples origin from the same individual.	Whole genome tumour short-read DNA sequencing datasets with >2.5% cross-patient contamination <i>should be rejected</i> .
Tumour contamination in germline samples	The proportion of tumour DNA present in germline sample, for example, for patients with haematological tumours where germline sample is collected after treatment.	Whole genome tumour short-read DNA sequencing datasets where matching germline sample contains >1% of tumour DNA <i>should undergo tumour-only analysis</i> .
Samples with low tumour purity	The proportion of tumour DNA in tumour sample.	Whole genome tumour short-read DNA sequencing datasets for samples with <30% tumour content <i>should be reported with warning</i> about reduced sensitivity of somatic variant detection.
Sample degradation	Composite metric that reflects the degradation of a tumour sample, and can be indicated by uneven coverage across AT-rich or GC-rich genomic regions, high percentage of chimeric reads and abnormal insert sizes.	Whole genome tumour short-read DNA sequencing datasets for degraded samples <i>should be reported with warning</i> about reduced sensitivity and specificity of somatic variant detection.
The metrics <i>shall</i> be considered in bioinformatics pipelines, and decisions made on whether to pass or fail samples. *The thresholds provided are guidance based on germline and tumour short read-based WGS approaches. Laboratories may benchmark and document their own thresholds for WGS, and alternative approaches, for example, whole exome sequencing and long-read sequencing, are expected to differ from the values presented in the table. WGS, whole genome sequencing.		

QC and assurance processes. Each of these major considerations is summarised in [table 4](#), and relevant case examples are provided in online supplemental case studies 1 and 2. These investigations include, but are not limited to, assessment against standard truth sets, assessment against specific clinical cohorts and external quality assurance schemes. In general, a minimal

value of 95% CI *shall* be determined for the performance of the approaches undertaken, except where explicit justification is provided, for example, CNV detection from formalin-fixed, paraffin-embedded (FFPE) tumour samples. The approaches adopted *shall* be deterministic, genetic variants detected *shall* be reproducible⁴² and this may be achieved by bioinformatics

Position statement

Table 3 Examples of findings from QC processes (see table 1) that may require mitigating actions during and/or after processing through the bioinformatics pipeline

Category	Indicative QC metrics	Suggested action
UPD	▶ Decrease in relatedness estimate to undertransmitting parent. ▶ Increase in relatedness estimate to overtransmitting parent. ▶ Localised increase in the number of Mendelian inconsistencies with undertransmitting parent. ▶ In cases of isodisomy, runs of homozygosity and genotypic differences to overtransmitting parent. ▶ In cases of heterodisomy, shared exact genotypes with the overtransmitting parent	Scientists/Clinicians evaluating the case <i>should</i> be alerted to the presence of a UPD, including reporting of relevant characteristics: ▶ Isodisomy/heterodisomy/mixed nature ▶ Segmental/complete chromosome involvement ▶ Overtransmitting parent
Large heterozygous CNV	Germline ▶ Runs of homozygosity. ▶ Possible increase in number of Mendelian inconsistencies. ▶ Possible alteration in relatedness estimates Mosaic ▶ Abnormal B-allele frequency distributions. ▶ Possible increase in number of Mendelian inconsistencies. ▶ Possible alteration in relatedness estimates.	If the bioinformatics pipeline is expected to identify large CNVs, no further action is required; high-quality and appropriate events should be identified and prioritised by the routine pipeline. If the bioinformatics pipeline does not identify and/or prioritise large CNVs, or mosaic CNVs, then clinical scientists/clinical team <i>should</i> be alerted to the potential presence of a CNV.
Unexpected familial relationships*	▶ Decrease or increase in relatedness estimate between proband and family member. ▶ If the relationship is proband-parent, then there is a widespread increase in the number of Mendelian inconsistencies.	Contact the referring team for approval to alter the referred relationship within the pedigree and/or remove the sample from downstream analysis strategies.

*May be indicative of incorrectly labelled samples, sample mix-ups or unexpected genetic relatedness between family members.
QC, quality control; UPD, uniparental disomy.

tools in isolation or through consensus-based approaches (eg, combining independent methods to detect structural variation). While the precise number of cases that are used to assess these areas of performance will differ across scenarios, specific recommendations *should* be adhered to where they are available, for example, germline structural variants.^{27 28 33}

Tertiary analysis approaches

Tertiary analysis refers to bioinformatics approaches that are undertaken downstream of the detection of genomic variation (online supplemental table 1).²⁴ This typically includes annotation of variants, and where appropriate, automated variant filtering and/or prioritisation strategies to triage genomic variants for further curation, review and classification. Strategies applied during the tertiary stage of the pipeline *shall* be consistent with appropriate variant interpretation and classification recommendations, and validation approaches *shall* be performed in partnership with appropriate stakeholders, for example, HCPC-registered clinical genomic scientists. Generally, tertiary analysis approaches *shall* aim to maintain high levels of sensitivity while ensuring the number of variants returned for interpretation is clinically manageable, in line with proportionate approaches to variant interpretation proposed in other guidelines.⁴³ For example, if the implementation of a new approach for *in silico* variant effect prediction increases the number of variants prioritised for clinical interpretation by 20%, then this needs to be assessed, with key stakeholders, to understand if the prioritisation approach remains proportionate and feasible given the specific clinical context of the testing strategy. These considerations are also applicable for pan-genomic markers, such as tumour mutation burdens, microsatellite instability, homologous recombination deficiency and other mutational signatures. Separate guidance is available for incidental findings⁴⁴ and bioinformatics pipelines *should* consider the latest guidelines and provide support for appropriate stakeholders. Furthermore, tertiary analysis approaches may be influenced by the family structures of the referral, for example, trios versus singleton

testing approaches.^{45 46} Where appropriate data are available, prioritisation strategies *shall* include familial disease and variant segregation patterns to help streamline variant interpretation. However, caution should be taken for such approaches to ensure that factors that may influence the sensitivity and diagnostic yield of the test are accounted for in analysis strategies, including uniparental disomies, interplay between disease presence and the possibility of allelic and genetic heterogeneity within the family, penetrance and age of disease onset.

A summary of the best practice recommendations for tertiary analysis approaches is included in table 4. The limitations of adopting strategies for annotation and prioritisation *shall* also be considered and clearly articulated to users. For example, if certain thresholds are considered, for example, allele frequency or scores from an *in silico* deleteriousness/pathogenicity tool, then reference *shall* be made to any appropriate benchmarking and/or recommendations that have been critical for these decisions. online supplemental case study 3 provides an example validation for upgrade to annotation resources that includes assessment of both sensitivity and specificity of the applied prioritisation approach.

A key consideration for tertiary analysis is the annotation resources used and the frequency with which they are updated. We recommend that, at a minimum, yearly appraisal of annotation resources *shall* be undertaken (table 4) and provide further advice where annotation resource updates *should* be considered more frequently than yearly (table 5). The integration of annotation resources *shall* consider both technical (eg, validate comprehensiveness of data and format) and analysis (eg, sensitivity and specificity for prioritisation) impact.

Statement of Limitations

To ensure transparency in clinical interpretation and reporting, a ‘Statement of Limitations’ *shall* be available for all cases processed by the bioinformatics pipeline. There are two distinct audiences for the statement of limitations: (1) HCPC-registered clinical genomic scientists and (2) clinicians

Table 4 Summary of best practice recommendations associated with secondary and tertiary analysis approaches

Category	Rationale	Best practice recommendation
Performance to detect genomic variants	The software and resources used in secondary analysis approaches have a direct impact on the performance to detect genomic variants.	► Strategies to assess performance <i>shall</i> include: (a) widely used truth sets, for example, Genome In A Bottle; (b) standard-of-care approaches and (c) external quality assurance schemes. ► Appropriate external quality assurance schemes to assess performance <i>shall</i> be participated in at least once annually, with findings clearly documented. ► Strategies to assess performance <i>shall</i> include all types of variants that are routinely included in the testing approach. ► All assessments <i>shall</i> include key measures of accuracy, including sensitivity, specificity and total number of errors, and where appropriate, assessments <i>should</i> include calculation of measurements of uncertainty and limits of detection.* ► Approaches adopted <i>shall</i> be deterministic, and genetic variants identified <i>shall</i> be reproducible.
Breadth and suitability of testing strategy	The software and resources used in secondary analysis approaches have a direct impact on the capability to appropriately survey genomic regions.	► Strategies <i>shall</i> assess that appropriate sequencing coverage is achieved for the genomic region included in current testing strategies.†
Precision of downstream processes	The software and resources used in secondary analysis approaches have a direct impact on downstream applications, for example, accuracy of variant prioritisation methods.	► The laboratory <i>shall</i> validate the impact that any change in software has on the precision of downstream variant analysis strategies, including assessment of the number of variants that are triaged for further clinical analysis. ► Any changes in behaviour <i>should</i> be discussed and approved by appropriate stakeholders, for example, service leads and/or clinical genomics scientists involved in running the service. ► Further consideration <i>should</i> be given to derived metrics from the bioinformatics pipeline, for example, alterations in behaviour of B-allele frequency estimations and impacts on other processes.
QC and assurance	The software and resources used in secondary analysis approaches have a direct impact on the distribution and suitability of thresholds applied in QC aspects of the bioinformatics pipeline.	► Thresholds used in QC approaches <i>shall</i> be validated against known truth sets, including revalidation after any change in software for secondary analysis approaches.
Variant annotation strategies	The methods adopted to annotate genomic variants can impact sensitivity, specificity and clinical utility of tertiary analysis approaches.	► At a minimum, there <i>shall</i> be a yearly appraisal of all annotation resources that are included in the variant annotation and prioritisation strategies. ► Some relevant annotation resources, for example, ClinVar, are updated frequently, and upgrades to tertiary analysis strategies <i>should</i> be actioned more frequently than yearly. See table 5 for additional best practice recommendations relating to variant annotation resources and suggested timeframes for evaluation.
Sensitivity and specificity of variant triage	The methods adopted to annotate genomic variants can impact sensitivity, specificity and clinical utility of tertiary analysis approaches.	► A validation set of variants, representative of the scope of the testing strategy, <i>shall</i> be agreed on in collaboration with appropriate stakeholders experienced in interpretation of variants for the type of test. In silico strategies to increase comprehensiveness of testing strategy <i>should</i> be considered. It is expected that multiple real-life cases with appropriate variant(s) present are required to validate new innovations and enable external quality assurance investigations. ► The sensitivity and specificity of the test <i>shall</i> be quantified and confirmed as clinically feasible prior to implementation.
Familial analysis strategies	Incorporation of familial variant and disease segregation patterns within analysis strategies can streamline clinical interpretation and increase diagnostic yields.	► Where appropriate data are available, prioritisation strategies <i>shall</i> include familial disease and variant segregation patterns. ► Factors which influence performance of familial analysis strategies <i>shall</i> be accounted for in analysis strategies, for example, uniparental disomy and considerations of variant penetrance.

*It is expected that appropriate Confidence Intervals are calculated for all assessed metrics.

†It should be noted that precise definition of 'appropriate sequencing coverage' is challenging, with a number of factors influencing the relationship between sequencing coverage and variant detection accuracy, including mappability of the region, quality/length of the sequencing reads and the exact variant detection algorithms adopted in the pipeline. Reduction in sequencing coverage for a given target region therefore may not represent a regression in performance.
QC, quality control.

referring patients to the diagnostic test. In both scenarios, the key objective is to ensure there is understanding of the technical limitations of the approaches undertaken, including types of genetic variants and genomic regions that are in or out of scope of the test. It may be appropriate to consider

separately written guidance for each stakeholder. The statement(s) *shall* clearly articulate and summarise the methods, limitations and quantified performance metrics for the applied pipeline. Examples of appropriate Statement of Limitations are provided in online supplemental case study 4 and online

Table 5 Summary of best practice recommendations for evaluating appropriateness of variant annotation data resources

Category	Best practice recommendation
All	<i>Shall</i> have a yearly appraisal, at a minimum, of all annotation resources and strategies for including these annotations in variant prioritisation and filtering. After the yearly appraisal, <i>shall</i> prioritise update resources and/or strategies in line with findings or establish appropriate risk mitigation approaches.
Transcripts	<i>Shall</i> use Matched Annotation from NCBI and EMBL-EBI (MANE) Select and/or MANE Plus Clinical transcript resources, as deemed appropriate for each gene, where these are available for the gene(s) in scope of the test May use additional transcripts, if required and justified.
Population frequency	<i>Shall</i> use gnomAD as a population frequency data resource, and <i>should</i> use the latest version of gnomAD. Alternative resources can also be used, with appropriate justifications. <i>Should</i> use in-house population frequency data derived for the same testing methods, or suitable alternative.
Population allele frequency*	<i>Shall</i> filter out common germline genomic variation from further prioritisation with a minimum allele frequency cut-off of 1%, unless specific recommendations explicitly state otherwise. <i>Should</i> use a stricter cut-off where appropriate to do so, for example, autosomal dominant genes.
Somatic allele frequency†	<i>Shall</i> use somatic frequency data resources to assess for enrichment of variants in previously characterised tumours. Examples of large-scale somatic variant resources include nationally curated databases, Catalogue of Somatic Mutations In Cancer (COSMIC) and Genie.
Disease databases	<i>Shall</i> use an inclusion list to rescue common (>1% allele frequency) disease-related variants. <i>Shall</i> share variant interpretation data between service providers where a mechanism is in place to do so. <i>Should</i> appraise the appropriateness of disease database resources used quarterly, for example, ClinVar.
In silico prediction tools	<i>Shall not</i> use in silico deleteriousness/pathogenicity prediction tools for filtering variants, unless specific recommendations explicitly state otherwise. <i>Shall</i> provide annotations from appropriate in silico tools, agreed with appropriate stakeholders, for example, Rare Exome Variant Ensemble Learner (REVEL), AlphaMissense, SpliceAI.
Prioritisation based on phenotype*	<i>Shall</i> use phenotypic terms (eg, human phenotype ontology) for prioritising disease relevant variants, where appropriate, for example, panel agnostic testing strategies.
Sequencing artefacts†	<i>Shall</i> use annotation resources to highlight technology-specific artefacts, for example, indels in homopolymer runs or repetitive regions for whole genome sequencing datasets. <i>Should</i> use an in-house panel of normal germline sequences derived for the same testing methods, or suitable alternative.

This guidance is applicable for all types of genetic variants within scope of the testing strategy, where appropriate resources are available.

*May not be applicable for strategies applied to tumour sequencing.

†May only be applicable for strategies applied to tumour sequencing.

supplemental case study 5. As guidance, at a minimum, the statement *shall* be easily understandable by an appropriate stakeholder and *shall* include: (1) a summary of the types of variants in scope and any associated and known limitations of these approaches along with measurements of uncertainty; (2) where appropriate, links to support documentation available for the user including dedicated user guides, standard operating procedures and any other sources of information that assists in understanding the methods and limitations of the bioinformatics pipeline; (3) clearly documented and validated accuracy of the bioinformatics pipeline, this *should* include validated limits of detection where appropriate, for example, for CNVs, short tandem repeat (STR) expansions, structural variants and mosaic variants, if considered in the pipeline and (4) where appropriate, a clearly defined scope of accreditation for the applied bioinformatics pipeline should be provided, with any aspects of testing outside the pipeline's accreditation scope clearly stated.

Development of software and data resources

Existing complications in the delivery of clinical high-throughput sequencing approaches include standardisation and sharing of software and data resources that are used across national service providers. There exist extensive and excellent resources that assist software developers and maintainers, and this document does not replicate them, including Github flow,⁴⁷ and other clinical bioinformatics best practice guidance.^{26 30–33 48} However, we highlight explicit best practice recommendations relating to software and data resource development, maintenance and sharing in [table 6](#), which are intended to sit alongside

these existing guidance. We mandate that bioinformatics software development *shall* adhere to best practices for clinical genomics services, with high-quality code and well-engineered systems as the goal, and *shall* be accredited to ISO 15189 and where appropriate, UKAS TPS71⁴⁹ and ISO 13485. Generally, development *shall* prioritise production of shared tools and resources, built around standards, with reusability and interoperability as key non-functional requirements.

Typically, clinical bioinformatics pipelines comprise secondary and tertiary analysis approaches that both augment and reference multiple data resources. These resources are also increasingly integrated with wider clinical systems such as laboratory information management systems and electronic health records. Bioinformatics data lend themselves to aggregation and subsequently being leveraged to improve patient care. Collaboration between laboratories is essential to maximise the impact of data sharing due to factors that include: complexity in measuring health outcomes; constrained breadth of individual teams (eg, lacking dedicated product and delivery managers); local expertise (eg, proficiency in assessing genome instability biomarkers, validating models for methylation signature profiling) and improvements to genomic analysis that result from standardised data sharing and learnings from wider cohorts. The guidance provided in the NHS digital service manual⁵⁰ and [table 6](#) addresses the increasing need for shared data resources across service providers and sits alongside national variant interpretation guidelines,⁵¹ which set out requirements for sharing secondary (variants) and tertiary (classification) data between service providers and

Table 6 Summary of best practice recommendations for software development, maintenance and data resource sharing

Category	Best practice recommendation
General	Bioinformatics software development <i>shall</i> adhere to best practices for clinical genomics services, with high-quality code and well-engineered systems as the goal. Best practices <i>shall</i> be applied to external software and resources. Where complete adherence is not possible, caution must be exercised regarding support, lifecycle and ongoing clinical suitability. Bioinformatics pipelines <i>shall</i> have version traceability and versions <i>shall</i> be clearly indicated in relevant documentation and reports.
Accreditation	Clinical bioinformatics services <i>shall</i> be accredited to International Organisation of Standardization standard 15189 (ISO 15189) and where appropriate, United Kingdom Accreditation Service Technical Policy Statement (UKAS TPS71) and ISO 13485. This is a requirement for all clinical laboratory services.
Software lifecycle	Validation and verification <i>shall</i> be coordinated by a change control process. This includes risk assessment, stakeholder approval and thorough documentation in a quality management system. Software testing strategies <i>shall</i> comprise unit, integration, regression and acceptance testing. The latter requires collaboration with appropriate stakeholders to ensure clinical utility and patient benefit. Cloud-first approaches <i>should</i> be considered for the development and ongoing innovation of bioinformatics approaches. Operating procedures for bioinformatics software <i>shall</i> acknowledge that genomics knowledge expands and develops and have clear plans for maintenance in order to stay up to date.
Programming standards	Bioinformatics software <i>shall</i> be developed collaboratively and <i>shall</i> prioritise production of shared tools and resources, built around standards, with reusability and interoperability as key non-functional requirements. Software <i>should</i> adhere to style guides for python, R, bash and SQL, and workflow description languages (eg, NextFlow and nf-core). Software <i>shall</i> be developed in line with other community standards including the Global Alliance for Genomics and Health (GA4GH) approved data formats and NHS data models, and with a shared understanding of good clinical governance.
Coding assistants	Strategies for the incorporation of AI coding assistants into bioinformatics software development and maintenance <i>shall</i> be developed in concert with broader healthcare strategies. There <i>shall</i> be careful adherence to appropriate information governance and data protection policies. AI coding assistants <i>shall</i> only be deployed alongside critical assessment of outputs.
Data sharing	Strategies for extraction and sharing of detected genetic variants, and associated clinical classifications, <i>shall</i> be implemented to ensure these data are easily accessible for the benefit of all NHS patients. Appropriate documentation <i>shall</i> be provided within software and README files. Shared code between service providers <i>shall</i> be containerised to increase portability and reproducibility.

AI, artificial intelligence; NHS, National Health Service.

therefore necessitate submission to national and public data repositories.

Alignment with existing guidance and recommendations

We compared recommendations developed through this work with those that have been published previously, including guidance that was published after the development and submission of this manuscript.³⁷ Overall, we compared 40 best practice recommendations described in this manuscript and presented in tables 1–6, identifying that 17 aligned with previous guidance, while 23 provided extra or novel information when considering information presented in all six publications that were used in the comparison. As each of the comparison papers tended to have a more restricted scope to single aspects of bioinformatics pipelines, we observed an expected high proportion of expanded and novel recommendations when performing pairwise comparisons (online supplemental table 3).

Of note, areas such as *performance to detect genetic variants* (table 4) are consistently provided as a recommendation across all literature used in the comparison, although the specific guidance provided here for the UK genetic testing ecosystem expands the information provided in other publications (categorised as ‘aligns’ with one publication and ‘expands’ against five publications). Notable areas where we provide extra or novel information compared with all six comparison publications include: specific comments on incidental findings from QC approaches (table 3); use of *in silico* prediction tools (table 5) and sharing data and experiences between laboratories and the use of programming assistants (table 6). We are highly supportive of the recommendations proposed elsewhere and note that the primary intention of this manuscript is to expand on the recommendations provided in 2016 with

a primary audience of bioinformaticians within the UK NHS and associated genomic diagnostic services. Comparisons with this guidance alone identify two recommendations that ‘align’, four recommendations that ‘expand’ previous guidance and 34 recommendations that are newly proposed in this manuscript.

DISCUSSION

Context

Key considerations of distributed national services for clinical genomic diagnostics are approaches to maximise equitability, reproducibility and innovation across service providers. In the UK, genomic diagnostic high-throughput sequencing has been, and will continue to be, applied across broad areas of national healthcare,⁵² including separate initiatives in England, Wales and Scotland. Bioinformatics approaches applied to assess data quality, to detect genomic variation, to prioritise and/or filter genomic variants for further clinical evaluation and to share data resources are core components of genomic testing strategies. Best practice recommendations for software development and application of clinical bioinformatics tools were published specifically for this purpose in the UK genetic testing ecosystem in 2016.³⁹ However, with the diversity of genomic sequencing approaches that currently exist within the UK NHS and the continued development of high-throughput sequencing chemistry and instrumentation,⁵³ consensus on best practice and frameworks for innovation and accreditation are essential to ensure standardisation to high-quality bioinformatics services across the UK. In this study, we update best practice recommendations through an ACGS-associated expert panel of bioinformaticians and scientists with experience across broad areas of genomics applied in diagnostics and healthcare, and compare those with other published

recommendations.^{24 25 33 36 37 39} We note that 40 essential best practice recommendations from this work are made available in [tables 1–6](#) and we direct readers to these resources, with text provided in the results and discussion to provide complementary overlap, further elaboration and context.

Scope of recommendations

The best practice recommendations proposed here include comments on primary, secondary, tertiary and QC aspects of bioinformatics approaches applied to clinical genomics (online supplemental table 1).²⁴ The guidance provided is highly supportive of recommendations outlined previously (online supplemental table 3)^{24 25 33 36 37 39} and is intended to sit alongside other existing guidance relating to variant interpretation and classification, for example, ACMG/AMP, ClinGen, ACGS, other associated variant interpretation and classification recommendations^{25 30 34 35 51 54–56} and the UK NHS National genomic test directory (which provides context for the scope of testing in different clinical scenarios and appropriate actual/virtual gene panels).⁵⁷ The guidance develops an innovation and accreditation framework for a broad range of DNA sequencing approaches, encompassing targeted enrichment and amplification approaches, and whole genome sequencing approaches adopted for germline and tumour genomic sequencing. Of note, these recommendations do not attempt to give prescriptive guidance for the exact thresholds or software tools used in specific testing scenarios, although to increase clarity and understanding, we provide specific examples in some sections, including supplementary case studies. For example, while the precise approaches that *shall* be applied for sequencing yield and coverage homogeneity/horizontal coverage are dependent on the exact testing approach, we provide example thresholds for germline and tumour genomic sequencing approaches ([table 2](#)) and note that germline whole genome sequencing is likely to have different QC thresholds when compared with whole exome sequencing, targeted gene panels and tumour sequencing. Moreover, different thresholds may be appropriate for the same general approach, for example, whole genome sequencing, across different sequencing technologies, such as long-read versus short-read approaches. As a result, we provide a general overview of the minimum requirements that *shall* be in place to ensure appropriateness of the sequencing datasets for clinical analysis ([tables 1 and 4](#)), and compare recommendations generated from our work with those that have been described previously in the literature (online supplemental table 3).

Cancer-specific considerations

A key distinction noted throughout the recommendations is areas of similarity and of difference between best practices applied in rare disease and in cancer diagnostic genomics. In general, due to the limited availability and the nucleic acid integrity of tumour samples, and the nature of tumour genomes, it may be difficult to assess and mitigate issues related to data quality. As a result, pass/fail criteria may not be pragmatic strategies, and laboratories *should* consider appropriate criteria to classify tumour sequencing datasets as 'low' or 'sufficient' quality, where appropriate. In addition to the QC metrics suggested for consideration in this regard ([table 1](#)), these assessments may include, but are not limited to, metrics that are indicative of DNA degradation (eg, coverage unevenness, average fragment sizes and extent of GC/AT dropout), and should be mindful of mutational processes in tumours that may impact performance (eg, loss of heterozygosity which may not be considered in contamination

assessment⁵⁸). Furthermore, we note clear caveats are required in certain scenarios, for example, a tumour sample may be classified as likely degraded ([table 2](#)) but may still provide the genomic answers to clinical questions, for example, presence of recurrent *POLE* exonuclease domain variants in endometrial cancer.^{59 60} It is noted that certain tumours are expected to cause clear differences in measurable characteristics, for example, *POLE* exonuclease domain variants may result in a hypermutated tumour genome with a massively increased number of variants detected, and these expected trends should be accounted for during QC assessments ([table 1](#)). The type of sample is also a key factor to consider in the QC thresholds used in the pipeline as DNA extracted from saliva or FFPE specimens may be more susceptible to DNA degradation than DNA extracted from peripheral blood or fresh frozen tissue.⁶¹ Finally, another challenge associated with cancer-specific analysis is the context in which the information from genomic testing is routinely integrated into electronic systems alongside concurrent investigations. While rare disease and germline analysis continues to evolve, this capacity and integration alongside phenotypic imaging is likely to become more widespread. There are current nationwide challenges and opportunities for bioinformatics approaches applied to cancer genomic sequencing in this context.

Potential clinical implications of best practice QC processes

Of note, it is possible that some of the best practice recommendations proposed for QC may identify findings that are appropriate for clinical reporting and/or require clinical input in order to decide best course of action. We recommend that processes *should* be in place that enable appropriate evaluation of cases that are detected from QC evaluations. Examples of appropriate scenarios and metrics that are indicative of these events are provided in [table 3](#) and include minor sex karyotypes and UPD, both of which may underpin genomic disorders.^{62 63}

Variant annotation resources

Another area of noted complexity during the creation of these best practice recommendations is the consistency of variant annotation resources across service providers, which may include internal and external resources, tools and databases. Variant annotation information can be used to automatically prioritise or filter variants for further consideration by appropriate stakeholders and/or be provided to assist with variant interpretation, classification and clinical reporting. As a result, alterations to annotation resources and/or strategies can impact the utility of the pipeline, for example, the choice of transcript resources may impact the predicted molecular consequence of a variant and thereby impact inclusion/exclusion of a variant for further curation and interpretation. We mandate the use of MANE Select and/or MANE Plus Clinical transcripts⁶⁴ for annotation, where these transcripts are available for genes included within the scope of the testing strategy. Moreover, the versions of datasets that are used for annotation are updated periodically, including population frequencies (eg, gnomAD releases⁶⁵) and disease variant databases (eg, ClinVar monthly updates⁶⁶). The outcomes from variant prioritisation strategies may differ by the exact versions of resources used in the pipeline, for example, a variant which is missing from an outdated version of the ClinVar database. Consistently updating databases and providing appropriate validation is challenging and through this study, we provide best practice recommendations in [tables 4 and 5](#) to assist appropriate and timely investigations that should be undertaken with regard to variant annotation. We give clarity on the use of

population frequencies, with preference for the use of the latest gnomAD version, and specify that the use of *in silico* deleteriousness/pathogenicity prediction tools *should* be used for annotation, but *shall not* be utilised for variant filtering unless other recommendations specifically mandate this. For example, recent guidance has suggested the use of a SpliceAI score threshold of 0.2 to filter specific categories of variants,³⁵ which is acceptable for incorporation into pipelines providing the limitations of such an approach are clearly articulated.

Scope and limitations

While this study provides best practices for bioinformatics applied to clinical genomics that covers broad scenarios and is applicable to multiple sequencing technologies, we also recognise that there are several types of high-throughput sequencing diagnostic tests that are out of scope of this guidance. For example, we do not attempt to provide guidance for RNA testing strategies and these recommendations have not intentionally considered epigenomic diagnostic testing, although it is expected that some of the proposed best practice recommendations will be relevant in this context. Future developments in this regard will be of high interest to the clinical genomics community. Moreover, the periodic assessment and update of best practice recommendations is required to ensure they remain appropriate for the rapidly evolving facets of clinical diagnostic testing facilitated by high-throughput sequencing. For example, current developments in genomics and bioinformatics indicate potential utility for clinical biomarker detection in expansive molecular profiling, including clinically actionable insights for cancer management and complex disease onset risk prediction. While clinical efficacy has yet to be fully understood, these confluences of multi-modal datasets and machine learning present a promising area for healthcare that *should* be closely monitored and considered as part of evolving best practices. Furthermore, we note the value of periodic reassessment and reanalysis of genomic datasets to identify new genomic diagnoses.^{67,68} While factors such as workforce capacity and appropriate automation currently limit its widespread adoption, we expect the development of clinical reanalysis and reannotation strategies to adhere to the appropriate frameworks proposed in this guidance in scenarios where they are developed. We recognise that across many of the challenges and issues raised, nationally adopted data standards and resources are critical for enabling the full potential of distributed NHS teams, and for the utilisation of clinical genomics to improve health. Such standards are not yet routinely adopted and we advocate for establishing national repositories for genetic variants and relevant clinical metadata, which can be interfaced with NHS infrastructure and information systems.

In summary, we consolidate viewpoints from across the UK to provide a clear set of best practice recommendations for clinical bioinformatics applied in the context of genomic sequencing. We anticipate that this guidance will be considered and implemented across broad areas of clinical genomics applied within the UK.

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